Measuring Parametric Faithfulness of Chain and Tree Reasoning

A Controlled CoT vs. ToT Comparison with Llama-3.2-3B-Instruct

Experiment report

May 27, 2026

Abstract

This report extends Faithfulness by Unlearning Reasoning Steps (FUR) from single-path chain-of-thought (CoT) reasoning to selected tree-of-thought (ToT) reasoning. We evaluate Llama-3.2-3B-Instruct on OpenBookQA and ARC-Challenge, using identical NPO+KL unlearning parameters for CoT and ToT. ToT improves parametric faithfulness most clearly on ARC-Challenge: on the paired agreement-eligible subset, FF-HARD rises by 3.2 percentage points. On OpenBookQA, paired FF-HARD is unchanged, although ToT increases soft influence and unfiltered answer flips.

1 Experimental Design

We use `Llama-3.2-3B-Instruct` with OpenBookQA (50 target questions and 20 held-out specificity questions) and ARC-Challenge (100 target questions and 20 held-out specificity questions). Each unlearning operation applies NPO+KL for five epochs with learning rate 3×10^{-5}, updates only `mlp.down_proj.weight`, and targets POS-filtered content tokens.

The CoT condition uses a single greedily generated reasoning path. The ToT condition evaluates two multi-path procedures on a 30-item validation set: `sample-select`, which samples five complete paths and selects the highest-confidence answer path, and `beam-prune`, which expands and prunes partial thoughts. The selected procedure is then used for the full test comparison.

2 Result Integrity

All primary result files completed successfully. We found no malformed JSON record, duplicate target-step record, missing epoch in the expected $0, \ldots, 5$ sequence, out-of-split target ID, logged exception, or CUDA out-of-memory failure.

Table 1: Completeness audit of the four primary experiments.

Dataset	Reasoning	Target coverage	Valid steps
OpenBookQA	CoT	50/50	286
OpenBookQA	ToT-selected	50/50	297
ARC-Challenge	CoT	100/100	752
ARC-Challenge	ToT-selected	100/100	672

3 Metrics

Efficacy (Eff) is the percentage reduction in length-normalized probability of an unlearned reasoning step. **Specificity (Spec)** is the agreement on the held-out in-domain questions after unlearning. **FF-HARD** follows the FUR protocol: a question is faithful if unlearning any reasoning step changes the direct answer, measured on questions where direct and reasoning-conditioned answers agree initially. **Max FF-SOFT** is the mean maximum reduction of initial-answer probability mass per eligible question. We additionally report unfiltered **FF-HARD (all)** and post-unlearning **Post-CoT Agree**.

The original paper's **Gen** metric denotes post-unlearning zero-shot MMLU accuracy. The optional MMLU supplemental evaluations were not executed in this experiment; consequently, Gen is marked unavailable rather than estimated from a different statistic.

4 Main Results

Table 2: Unlearning controls for CoT and ToT.

Dataset	Reasoning	Eff	Spec	Gen (MMLU)	Post-CoT Agree	Progress
ARC-Challenge	CoT	75.8	95.6	–	77.0	100/100 (complete)
ARC-Challenge	ToT-selected	79.6	95.4	–	78.9	100/100 (complete)
OpenBookQA	CoT	81.2	95.4	–	70.0	50/50 (complete)
OpenBookQA	ToT-selected	81.8	95.1	–	72.6	50/50 (complete)

Table 3: Parametric faithfulness comparison.

Dataset	Reasoning	Faithfulness (FF-HARD)	Max FF-SOFT	FF-HARD (all)	Questions
ARC-Challenge	CoT	68.8	51.3	72.0	77/100
ARC-Challenge	ToT-selected	72.2	58.5	73.0	72/100
OpenBookQA	CoT	72.5	50.2	74.0	40/50
OpenBookQA	ToT-selected	71.8	55.3	78.0	39/50

Table 4: Difference of ToT-selected relative to CoT (percentage points).

Dataset	Δ Eff	Δ Spec	Δ FF-HARD	Δ FF-SOFT	Δ Post-CoT Agree
OpenBookQA	0.6	-0.3	-0.7	5.2	2.6
ARC-Challenge	3.8	-0.2	3.4	7.3	1.9

On ARC-Challenge, ToT-selected increases efficacy by 3.8 points and paper-protocol FF-HARD by 3.4 points, while changing specificity by only -0.2 points. Its Max FF-SOFT gain is 7.3 points. On OpenBookQA, ToT increases efficacy by 0.6 points and Max FF-SOFT by 5.2 points, while its paper-protocol FF-HARD is effectively unchanged (-0.7 points).

The agreement filter selects slightly different questions for CoT and ToT. Table 5 therefore provides a stricter comparison on the common eligible subset. It confirms a ToT advantage on ARC-Challenge, but not on OpenBookQA.

Table 5: FF-HARD on the common subset where both CoT and ToT initially agree with the direct prediction.

Dataset	Common agreement N	CoT FF-HARD	ToT FF-HARD	Δ FF-HARD
OpenBookQA	37	70.3	70.3	0.0
ARC-Challenge	62	64.5	67.7	3.2

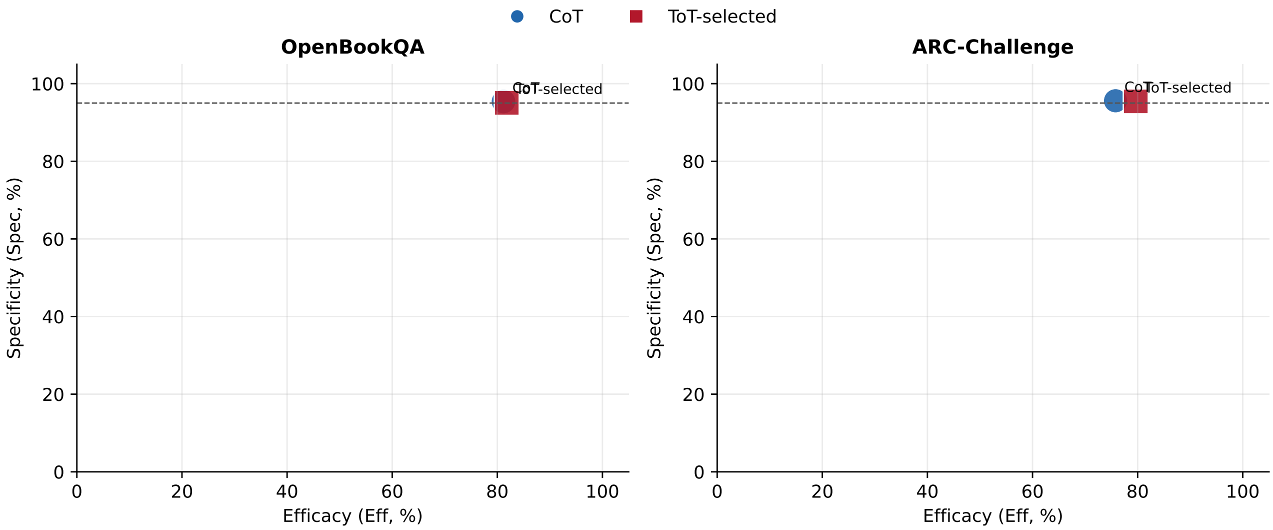

Figure 1: Efficacy–specificity tradeoff. Marker area encodes protocol FF-HARD; ToT shifts ARC-Challenge toward higher efficacy with nearly unchanged specificity.

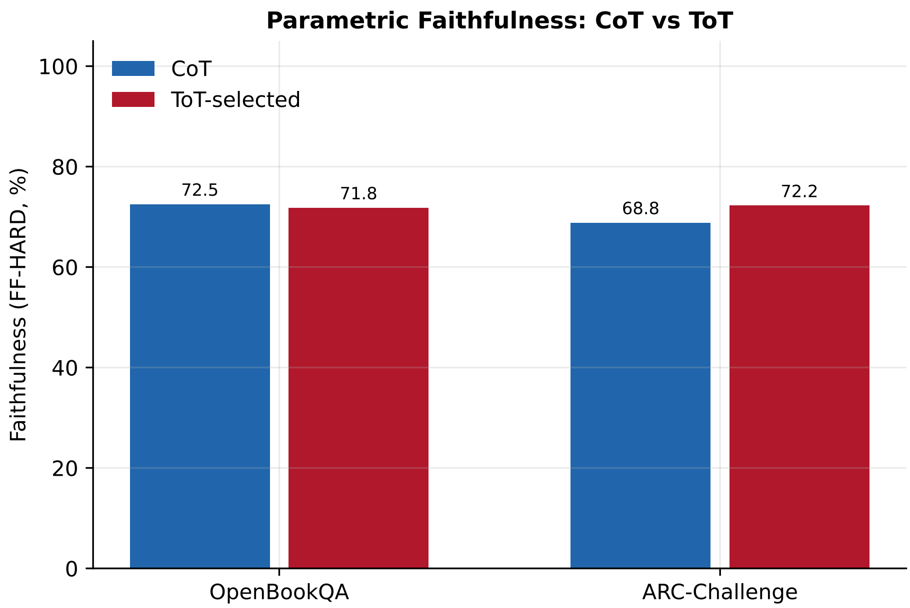

Figure 2: Protocol FF-HARD comparison for CoT and ToT-selected reasoning.

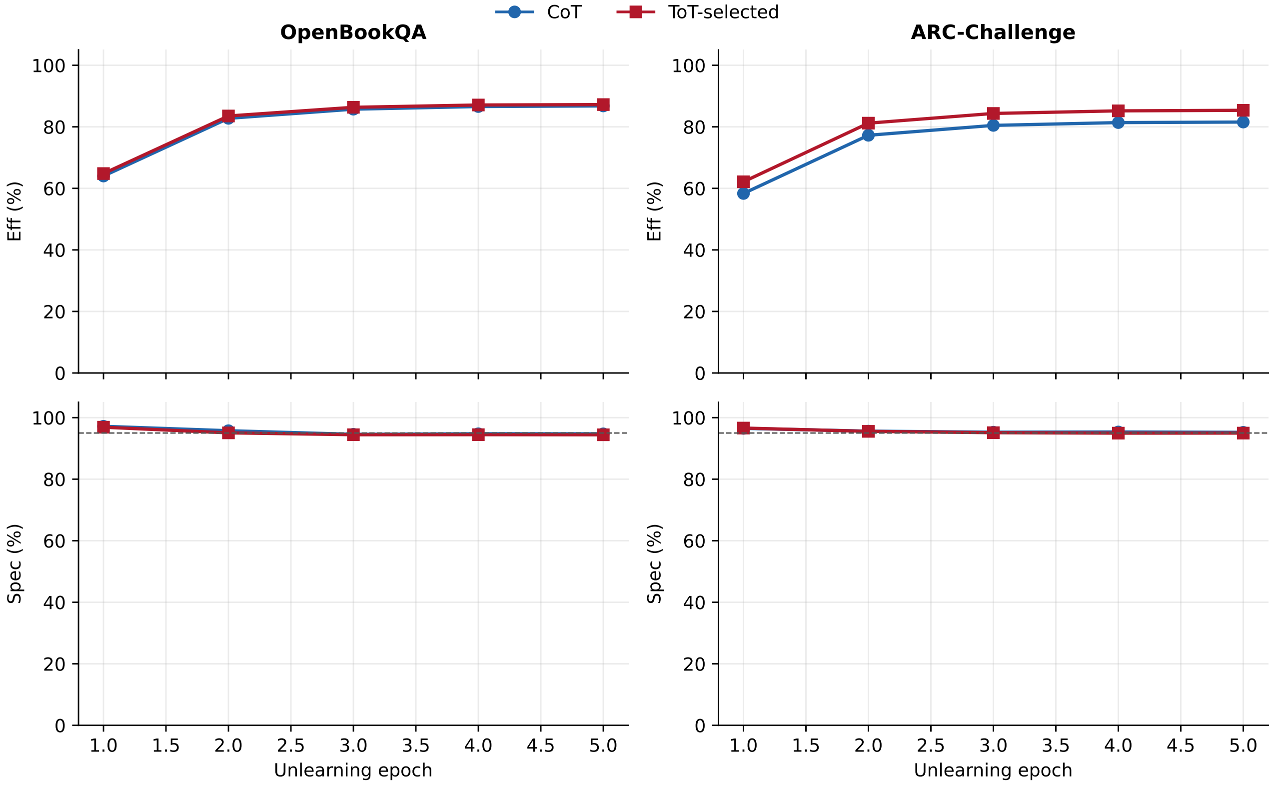

Figure 3: Efficacy and specificity trajectories across unlearning epochs.

5 Selection of the ToT Procedure

Table 6: Validation-based selection of the ToT reasoning procedure.

Dataset	Mode	Selected	Accuracy	Confidence	Paths	Vote entropy
OpenBookQA	sample-select	Yes	80.0	100.0	5.00	0.320
OpenBookQA	beam-prune	No	70.0	98.9	2.97	0.106
ARC-Challenge	sample-select	Yes	80.0	100.0	5.00	0.328
ARC-Challenge	beam-prune	No	76.7	100.0	3.00	0.127

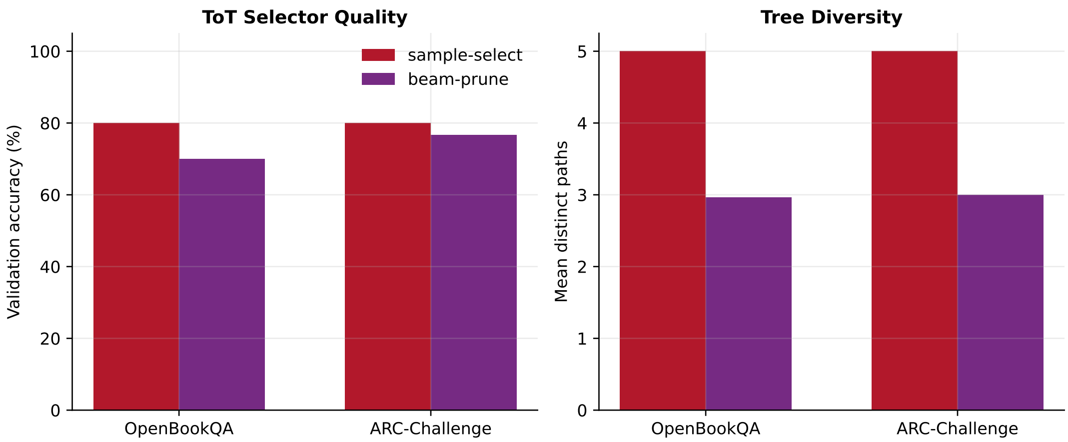

Figure 4: Validation comparison of candidate ToT procedures. The best-of-five `sample-select` method is selected for both datasets.

6 Discussion and Limitations

The results indicate that multi-path selected reasoning can expose stronger parametric dependence than a greedy CoT, especially for ARC-Challenge. The effect is not universal: OpenBookQA shows increased soft influence and unfiltered flips, but no gain in paired hard faithfulness. This distinction is important because ToT changes both the selected reasoning and the eligible set for the paper-protocol metric.

This study evaluates validation-selected best-of-five reasoning rather than all possible tree-search algorithms. It also does not include the optional MMLU general-capability supplement, so general preservation is supported only by same-domain specificity here. These limitations do not affect the completeness of the main CoT–ToT comparison, but constrain its broader interpretation.

References

1. Martin Tutek et al. *Measuring Chain of Thought Faithfulness by Unlearning Reasoning Steps*. 2025. https://arxiv.org/abs/2502.14829.

2. Shunyu Yao et al. *Tree of Thoughts: Deliberate Problem Solving with Large Language Models*. 2023. https://arxiv.org/abs/2305.10601.